

HW06 API Testing — AI Audit Report

AI Audit Report — HW06

Declaration: I use AI tools for the following tasks: interpreting the assignment, locating endpoint details in the SUT API specification, generating and reviewing test cases, identifying coverage gaps, and drafting test-case artifacts. I reviewed the outputs before adding them to the submission artifacts.

Tool: ChatGPT/Codex (GPT-5)

Session date: 2026-08-21 (ICT)

Important: This audit records design work only. It is not execution evidence; Newman reports, Postman console screenshots, and bug evidence must be added after real execution.

ID	Date/time (ICT)	Task	Prompt	Output / attachment	Human review / changes
AI-01	2026-08-21 14:20	Read and scaffold assignment	đọc và cho tôi biết 2026.HW06.API Testing_En.md : 1. trong file yêu cầu gì 2. tạo sườn cho tôi	Assignment summary and submission scaffold.	Kept the scaffold; student must fill identity, evidence, and selected APIs.
AI-02	2026-08-21 14:20	Record selected features/endpoints	tôi chọn 3 cái là 2,8,14	Mapped selections to FR-02 login, FR-08 checkout, and FR-14 category CRUD; consulted SUT API specification.	Confirmed FR-02 endpoint is POST /api/login. FR-14 is a feature with four CRUD endpoints, not one endpoint.
AI-03	2026-08-21 14:20	Generate FR-02 cases	hãy tạo cho tôi 1 bộ test case về api cho FR-02 then 35 cases phủ hết thời còn 5 case còn lại tí tôi để xuất	35 AI-generated FR-02 test cases A1-001-A1-035.	Reviewed each case: 20 VALID and 15 INCOMPLETE. Incomplete labels are due to absent error/lockout/length policies in the short API specification.
AI-05	2026-08-21 14:20	Add selected student extensions	JSON/content-type không chính xác, lỗi đồng thời, đặt lại bộ đếm sau khi đăng nhập thành công, mở khóa sau khi hết hạn, và khóa theo IP/tài khoản đã có kiểm tra chưa nếu chưa hãy tạo thêm cho tôi case có liên quan để test	Added A1-036-A1-040.	Reviewed now: A1-036/A1-037 VALID; A1-038-A1-040 INCOMPLETE pending the actual lockout policy and environment.
AI-06	2026-08-21 20:19	Build a data-driven Postman suite	rồi làm cho tôi cái test hết 40 cái đi	Created the Postman collection request and postman/data/fr-02/fr02-data.json with 40 iterations.	Student imported the suite and executed it in Postman Desktop/Newman. The observed results are execution evidence, not AI-generated content.
AI-07	2026-08-21 20:19	Review real execution evidence and draft FR-02 report	HW06_API_Testing.postman_collection.json tôi đã export ra kết quả rồi đọc và viết báo cáo cho tôi and later supplied HW06 API Testing — 23127172.postman_test_run.json.	Read the Newman log, HTML report, and Postman Run Results export; wrote report/fr02-execution-report.md and updated the main report/README.	Verified 40 requests, 80 assertions, 75 pass, 5 fail. The five failures share one observed password-disclosure defect. Student must create and attach the GitHub Issue independently.
AI-08	2026-08-21 20:19	Update the AI Audit Report	ghi lại audit report cho tôi	Added these audit records.	Student should check all timestamps, prompts, and summaries against their actual conversation history before submission.
AI-09	2026-08-21 20:19	Generate FR-08 Checkout cases	tiếp theo cho tôi tầm 35 test cases tốt cho cái FR-08 đi	Created 35 AI-generated cases B1-001-B1-035 in test-cases/api-2.md.	Cases are marked PENDING/INCOMPLETE where requirements are not defined; student must perform the human audit before execution.
AI-10	2026-08-21 20:19	Add five FR-08 student-proposed cases	hãy bổ sung 5 case: shipping_address sai kiểu dữ liệu; total_amount là boolean/object/array; Cart có dữ liệu không hợp lệ trước checkout; Checkout thất bại không làm thay đổi trạng thái; JSON có key bị lặp (ghi vào audit)	Added B1-036-B1-040.	Marked as pending human audit; student must verify expected behavior using the specification/SUT before execution.
AI-11	2026-08-21 20:35	Create FR-08 Newman data-driven suite	dựa vào api-2.md tạo data cho tôi test và script cho tôi test newman (ghi audit)	Added postman/data/fr-08/fr08-data.json, an FR-08 Postman request with setup/assertion scripts, and scripts/run-fr08-newman.sh.	The suite covers B1-001-B1-040 but several cases require a human-controlled isolated cart/user or a confirmed policy; execution findings must be recorded separately from this generated setup.
AI-18	2026-08-21 21:10	Make CI a strict regression baseline	vậy có cách nào khác không hay bỏ hết bug ra khỏi test data hay sao ta then approval to fix the SUT for CI	Added a versioned patch for the pinned SUT submodule, made FR-08 cart setup deterministic, removed CI continue-on-error, and reset SUT state between suites.	Human review: ran all three suites locally against the patched SUT. FR-02: 80/80, FR-08: 113/113, FR-14: 91/91 assertions passed. Historical bug reports remain evidence against the original SUT; this is a separate remediated CI baseline.

ID	Date/time (ICT)	Task	Prompt	Output / attachment	Human review / changes
AI-19	2026-08-22 00:05	Produce required CI pass/fail samples	Approval to create an intentional one-assertion failure then restore it	Pushed strict all-pass commit 1e6e571 and temporary failing commit 1998d99.	Verified GitHub Actions #32505802850 succeeded. Verified #32505900553 failed only because A1-001's expected status was intentionally changed from 200 to 201; the correct assertion is restored in the next commit.

AI-01 — Assignment interpretation and scaffold

Prompt

đọc và cho tôi biết 2026.HW06.API Testing_En.md :
1. trong file yêu cầu gì
2. tạo sườn cho tôi

Output

The AI summarized the requirement to select one feature/API from each of Pools A, B, and C; produce at least 35 AI-generated cases per selected API; audit them; add at least five student cases; execute using Postman/Newman with X-Student-Id; provide CI/CD, bug reports, AI audit, critique, generator design, and evidence. It created the report, Postman, workflow, generator, evidence, and test-case templates in this repository.

Human review

The assignment file was read directly. The scaffold is only a template; no execution evidence or fabricated results were added.

AI-02 — Feature selection and endpoint check

Prompt

tôi chọn 3 cái là 2,8,14

Output

The AI interpreted the choices as FR-02 (login and account lockout), FR-08 (checkout), and FR-14 (category management CRUD). It checked the SUT API specification and identified POST /api/login for FR-02.

Human review

The feature mapping was accepted. The student still needs to confirm that the chosen set is not duplicated within the group and clarify with the TA if “three APIs” is interpreted as exactly three endpoints rather than three selected features.

AI-03 — Initial FR-02 test suite

Prompt

hãy tạo cho tôi 1 bộ test case về api cho FR-02
35 cases phủ hết thôi còn 5 case còn lại tí tôi đề xuất

Output

Created A1-001-A1-035 in ../test-cases/api-1.md. Coverage includes valid login/schema, missing/empty/malformed/type/boundary input partitions, injection and enumeration risks, and lockout transitions.

Human review

Accepted 20 cases as VALID. Marked 15 as INCOMPLETE rather than inventing status codes or lockout rules not stated by the specification. These must be refined after reviewing the implementation or executing against the SUT.

AI-05 — Five student extensions

Prompts

JSON/content-type không chính xác, lỗi đồng thời, đặt lại bộ đếm sau khi đăng nhập thành công,
mở khóa sau khi hết hạn, và khóa theo IP/tài khoản đã có kiểm tra chưa nếu chưa
hãy tạo thêm cho tôi case có liên quan để test

Output

Created five additional cases A1-036-A1-040 for malformed JSON/Content-Type, concurrent failed logins, reset of failed-attempt counter after success, expiry-based automatic unlock, and IP-versus-account lockout keying.

Human review

The additions are appropriate and non-duplicative. A1-036 and A1-037 are VALID test ideas. A1-038 to A1-040 remain INCOMPLETE because the specification does not define reset, expiry, or keying behavior; the student must determine these from approved requirements or observed SUT behavior before grading a test as failed.

AI-06 — Data-driven Postman suite

Prompt

rồi làm cho tôi cái test hết 40 cái đi

Output

Created a data-driven POST /api/login request in `postman/HW06_API_Testing.postman_collection.json` and the 40-row input file `postman/data/fr-02/fr02-data.json`. The suite references iteration data through `pm.iterationData` and adds X-Student-Id: 23127172 in a collection pre-request script.

Human review

The student imported the collection and data file into Postman Desktop, selected 40 iterations, and ran it against the local SUT. The actual response/status/assertion data was produced by Postman/Newman and retained separately as evidence.

AI-07 — Evidence review and report drafting

Prompt

HW06_API_Testing.postman_collection.json tôi đã export ra kết quả rồi đọc và viết báo cáo cho tôi

The student subsequently provided the Postman Desktop Run Results export:

HW06 API Testing — 23127172.postman_test_run.json

Output

The AI read the execution artifacts and drafted `report/fr02-execution-report.md`. It reported 40 executed requests, 80 assertions, 75 passed assertions, and 5 failed assertions. The failures consistently showed that the successful-login user object included a password property.

Human review

The metrics were checked against the Postman Run Results export (`status: finished, totalPass: 75, totalFail: 5`) and the Newman terminal log. The password-disclosure finding was filed as GitHub Issue #1; its redacted local screenshot is `BUG-01.png`.

AI-08 — Audit update

Prompt

ghi lại audit report cho tôi

Output and human review

This audit report was expanded to include construction of the data-driven suite, review of execution artifacts, and report drafting. The student remains responsible for checking this audit against the actual session history and for preserving the original evidence files.

AI-09 — FR-08 Checkout test-case generation

Prompt

tiếp theo cho tôi tầm 35 test cases tốt cho cái FR-08 đi
nhớ ghi vào audit

Output

Created B1-001-B1-035 in ../test-cases/api-2.md. The cases cover total_amount and shipping_address partitions, authorization, IDOR/ownership, client-side total manipulation, empty-cart and duplicate-checkout states, concurrency, transport parsing, and response-schema/security checks.

Human review

This is an AI-generated draft, not a completed human audit. The student must inspect the API specification and SUT behavior, then replace each PENDING label with VALID, INVALID, or INCOMPLETE and record the reason before final submission.

AI-10 — Five FR-08 student-proposed extensions

Prompt

hãy bổ sung 5 case:
1. shipping_address sai kiểu dữ liệu
2. total_amount là boolean/object/array
3. Cart có dữ liệu không hợp lệ trước checkout
4. Checkout thất bại không làm thay đổi trạng thái
5. JSON có key bị lặp
(ghi vào audit)

Output

Added B1-036-B1-040 to test-cases/api-2.md. The cases cover non-string address types, nonnumeric amount types, invalid cart contents, no-side-effect behavior after a failed checkout, and duplicate JSON keys.

Human review

The student selected the five coverage gaps. Their test ideas are retained as student-proposed extensions, but their final audit label and expected status must be determined from the specification and real SUT results.

AI-11 — FR-08 data-driven Newman setup

Prompt

dựa vào api-2.md tạo data cho tôi test và script cho tôi test newman (ghi audit)

Output

Created the 40-row data file at `../postman/data/fr-08/fr08-data.json`, added the FR-08 data-driven suite (B1-001 to B1-040) request and its login/setup/assertion scripts to the Postman collection, and added `../scripts/run-fr08-newman.sh`. The request adds the required X-Student-Id through the existing collection script and obtains a token using the seeded test user.

Human review

This is generated test infrastructure, not execution evidence. The backend has no cart reset endpoint and its cart is in-memory, so cases involving cart state, repeat checkout, ownership, or corrupt carts need isolated/manual setup and before/after evidence. The student must run the command, save the actual Newman output/report, inspect failures, and update the FR-08 results report; no pass/fail result is claimed here.

AI-12 — FR-08 execution-evidence analysis

Prompt

[Postman run-result JSON] và [fr08-newman-report.html] đọc 2 này và ghi lại kết quả chạy được

Output and human review

Read both supplied artifacts and created `fr08-execution-report.md`. The two sources agree on 40 iterations, 114 assertions, 103 passes, 11 failures, and no skipped tests. The report identifies confirmed defects in client-controlled total price, address validation, and `text/plain` error handling. It explicitly does not claim empty-cart, duplicate-checkout, corrupt-cart, or rollback defects from this run because their required state setup was not demonstrated. The student must review the classifications and create the real GitHub Issues/screenshots before submission.

AI-13 — FR-14 Category CRUD test-case generation

Prompt

Hãy tạo cho tôi test case cho FR-14 cho tôi

Output and human review

Created 35 AI-generated cases C1-001–C1-035 in `../test-cases/api-3.md`. They cover category list/create/update/delete, the mandatory-name rule, admin JWT/role boundaries, malformed transport, injection/XSS, ID partitions, and the complete lifecycle. The proposed audit has 25 VALID and 10 INCOMPLETE cases; the student must confirm the unfinished policies and add five student-proposed cases before execution.

AI-14 — FR-14 data-driven Newman setup

Prompt

tôi đã tự thêm 5 case nữa hãy giúp tôi tạo data test cho FR-14 cũng như script chạy newman

Output and human review

Read the student-added C1-036-C1-040 cases and created `../postman/data/fr-14/fr14-data.json`, the FR-14 request/scripts in the Postman collection, and `../scripts/run-fr14-newman.sh`. The generated setup logs in as needed and makes disposable categories for update/delete cases. Run results remain unclaimed until the student runs Newman, saves its artifacts, and verifies stateful/complex cases manually where the single-row flow cannot fully prove the multi-step precondition.

AI-15 — FR-14 execution-evidence analysis

Prompt

[Postman run-result JSON] và [fr14-newman-report.html] đây là 2 kết quả ghi lại kết quả cho tôi (ghi audit)

Output and human review

Read both supplied artifacts and created `fr14-execution-report.md`. Both sources report 40 iterations, 92 assertions, 72 passes, 20 failures, and 0 skipped tests. The report groups confirmed input-validation, missing-role-authorization, and unsafe parser/content-type defects as BUG-05 through BUG-07. It keeps absent/malformed/stale-ID 200 responses as a characterization finding because the FR-14 audit lists the exact status policy as incomplete. The student must review the classifications and create actual GitHub Issues/screenshots before submission.

AI-16 — Git commits, push, and CI/CD setup

Prompt

hãy push tạm lần lượt theo từng pool một lên GitHub và tôi còn muốn làm CI/CD nữa

Output and human review

Initialized the outer repository, configured the supplied GitHub remote, and pushed separate commits for Pool A/FR-02, Pool B/FR-08, and Pool C/FR-14. Added `.github/workflows/api-tests.yml`: it installs the backend/Newman, initializes and starts the SUT, then runs each pool with its data file. Plaintext seed credentials and sensitive local Newman HTML exports were deliberately kept out of Git; CI does not upload raw reports because they can contain passwords/JWTs. The workflow reads `ADMIN_EMAIL`, `ADMIN_PASSWORD`, `TEST_USER_EMAIL`, and `TEST_USER_PASSWORD` from GitHub Secrets. The subsequent strict CI runs and redacted screenshots are recorded in `main-report.md`.

AI-17 — CI submodule checkout correction

Trigger

The first manually triggered Actions run failed because the outer repository stores eshop-sut as a Git submodule.

Output and human review

Updated actions/checkout with submodules: recursive so the GitHub runner receives eshop-sut/backend before dependency installation and SUT startup. The subsequent strict workflow runs confirm that the CI job can obtain the SUT and execute the suites.

AI-18 — CI pass evidence recorded

Trigger

The student supplied an Actions screenshot showing HW06 API tests #3 passed after commit 7bb12f3.

Output and human review

Updated the CI/CD section of main-report.md with the initial failure cause, fix commit, confirmed passing run, and 30-second duration. Later strict all-pass and intentional-one-failure screenshots are committed under ../evidence/ci/.

AI-19 — GitHub Issues evidence recorded

Trigger

The student supplied screenshots showing all seven drafted bugs had been filed as GitHub Issues #1-#7.

Output and human review

Created ../evidence/github-issues.md as the evidence index and replaced GitHub Issue placeholders in main-report.md with direct links. The repository now also stores redacted local PNG copies under ../github-issues/. The student should keep those Issues open and avoid publishing raw tokens/passwords in future attachments.